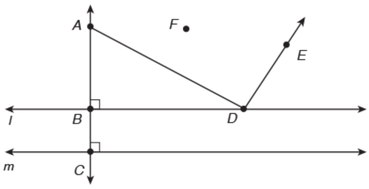
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Problem Solving with Angles

Geometry is concerned with the properties and measurement of points, lines, angles, surfaces, and solids. The first step in learning geometry is to understand its terms. Refer to the drawing as you review the definitions on this page.



A **point** is a precise location in space. The drawing has six labeled points: A through F.

A **line** is a straight path of points that continues in opposite directions. A line has infinite length shown by an arrow on each end. A line can be named using a single italic letter or by two points on the line.

Line *l* can also be called line *BD*, also written $\overleftrightarrow{BD}$.

A **ray** is a straight path of points that starts at one point and continues infinitely in one direction. Ray *DE*, also written $\overrightarrow{DE}$, starts at point *D*, passes through point *E*, and continues.

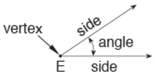
A **line segment** is the path of points between two definite points. A line segment is named by its endpoints. Find segment *AD*, also written $\overline{AD}$.

Some lines belong in special categories. **Vertical** lines run straight up and down. **Horizontal** lines run left and right. In the drawing, lines *l* and *m* are horizontal and $\overleftrightarrow{AC}$, the line passing through points *A* and *C*, is vertical.

Two lines **intersect** when they cross at a point. Lines *AB* and *BD* intersect at point *B*. Some lines will never intersect. Two lines that run in the same direction are called **parallel** lines. No matter how far parallel lines are extended, they will never cross. In the drawing, lines *l* and *m* are parallel.

Lines that intersect to form square corners (also called right angles) are called **perpendicular** lines. $\overleftrightarrow{AC}$ is perpendicular to lines *l* and *m*.

A **vertex** is a point where two rays or line segments meet or intersect. An angle is formed by two rays. The rays that form an angle are called the sides of the angle.

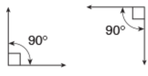
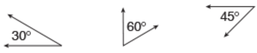
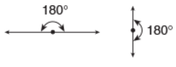




The angle increases as the hand moves.

The size of an angle depends on the amount of **rotation** of the sides. Imagine a clock with both hands pointing at the 12. As one hand moves clockwise, the measure of the angle created by the hands increases. Angles are measured in degrees (°). One full rotation (forming a complete circle) is 360°.

Below are the properties of four angles that you will need to know.

Name of Angle	Properties	Examples
right angle	exactly 90°	
acute angle	less than 90°	
straight angle	exactly 180°	
obtuse angle	between 90° and 180°	

Key Terms

- adjacent angles
- complementary angle
- intersect
- line
- point
- polygon
- ray
- line segment
- supplementary angle
- vertical angles
- vertex

Tip Hide

A right angle is indicated by a small box drawn inside the angle.

Rate Your Confidence

High
Medium
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Continue

Category

Lesson 1: Basic Figures

Related Questions:

Practice questions taken related to this topic

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